

ConGAT: Context-Aware Graph Attention for 3D Region Discovery in Multiplexed Microscopy

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Abstract—The global incidence of cancer is increasing, underscoring the growing importance of early detection. Understanding the spatial distribution of tumor and immune cell subsets in suspicious tissues is essential for researchers studying the progression of carcinogenesis and the effects of treatment. New multichannel microscopic imaging methods, such as cyclic immunofluorescence (CycIF), are helpful for precisely and detectably mapping biomarkers at the single-cell scale in tissue biopsies.

Researchers often analyze 2D microscopic images with AI, but current methods rely heavily on time-consuming human annotations and ignore 3D information. This limits biological insights and the identification of ROI. To address this, we propose ConGAT, a 3D graph attention network (GAT) that uses self-supervised refinement of positional, morphological, and intensity features to identify meaningful ROIs in 3D microscopy data. It removes the requirement for labeled data, utilizes depth information frequently obscured in 2D images, and generates 3D spatial graphs from CycIF images. The model develops ROI representations that account for both spatial and biological contexts, allowing for automatic detection of important areas that other methods might overlook because of clutter, occlusion, or tissue heterogeneity. Its flexible design enhances performance across various datasets and microenvironments.

Index Terms—Healthcare Informatics, Graph Attention Network (GAT), Self-Supervised Learning, Multiplexed Microscopy Imaging, Immune-Tumor Microenvironment, Visual Analytics, Region of Interest (ROI).

I. INTRODUCTION

Recently, high-dimensional and multi-channel microscopic imaging has become a central component of modern disease diagnosis and biomedical research, supporting detailed consideration of tissue structure and cellular organization [1], [2]. Multiplexed imaging techniques, like t-CyCIF [3] and CODEX [4], enable detailed, high-resolution profiling of biomarker populations within tissue samples. They generate high-dimensional data concerning biomarker expression, morphology, and intensity. Additionally, these methods provide contemporary insights into tissue-level disease processes and the organization of the microenvironment [3], [5]. A range of computational approaches, including convolutional neural networks and graph-based models, have been applied to analyze such data by identifying biomarker distributions and their interactions [6], [7]. However, most available techniques work with two-dimensional representations and are highly dependent on manual annotations, which restricts their capacity to represent intricate three-dimensional spatial dependencies [8]–[10]. Additionally, autoencoders and GATs have demonstrated

the capability to capture adaptive cellular interactions at a large scale [11], [12]. These approaches typically rely on labeled datasets, which are expensive and often difficult to collect in real-world biomedical studies; thus, they will restrict most supervised machine learning strategies [13]–[15]. Still, this lack motivates the development of self-supervised methods capable of learning intrinsic spatial and molecular representations directly from data [16], [17]. These developments inherently require self-supervised graph attention models, as they may help detect biologically relevant structures in complex, high-dimensional tissue microenvironments [13], [18]. Nevertheless, current clustering approaches, 2D CNNs, and graph frameworks are still incapable of capturing the volumetric spatial organization and cell-cell interactions that characterize the tumor-immune microenvironment [6], [19]–[21]. These limitations are further compounded by the lack of integration with interactive visualization systems, making them difficult to interpret biologically and to validate using expert knowledge [22]–[24]. The solution to these gaps is the development of unsupervised, interpretable structures that work directly on multi-channel 3D CycIF data, facilitating both quantitative analysis and discovery [3], [17], [18], [25]. To overcome these limitations, we introduce ConGAT, a self-supervised pipeline of GATs capable of detecting biologically significant ROIs, like tumor-immune niches, within large-scale, high-dimensional, multiplexed tissue microscopic images. Our method represents each supervoxel as a graph node that includes biomarker intensities, morphological features, and spatial coordinates. Also, Edges are features that encode possible relationships based on spatial distance and phenotypic similarity, thereby providing a three-dimensional texture structure. The model dynamically uses attention mechanisms on multi-layered graphs to weight inter-node relationships based on their noise-reduction effects and emphasis on biologically salient interactions. ConGAT learns the complexities of biomarker co-expression and scale-dependent spatial organization through self-supervised learning, and without the use of any labeled data. In addition to its analysis function, ConGAT will also generate interpretable results and visualize them using an interactive visualization framework, such as Vitessce [24]. The method empowers the domain experts to explore areas of detection, make model discoveries, and rationalize spatial interaction patterns in the context of the tissue. ConGAT will help with quantitative modeling, expert interpretation, and integration by consolidating the features.

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These features allow for scalable and interpretable analysis of the intricate tumor-immune microenvironment, laying the groundwork for extensive biomedical research and possible clinical uses.

II. RELATED WORK

Current computational and visual analytics methods have enhanced the study of spatially resolved and multiplexed tissue imaging measurements [3], [9]. However, they are still limited in their ability to capture biologically significant interactions within a three-dimensional microenvironment. Recent algorithms, such as neural networks, autoencoders, and graph-based spatial GNNs, as seen in STAGATE, SpaGCN, and SPACE-GM [6], [14], [20], [21] have been widely applied to multiplexed image datasets to extract features and neighborhood structures. However, most graph formulations treat inter-marker relationships uniformly, limiting their ability to distinguish biologically meaningful interactions from noisy or redundant connections. To overcome this weakness, our approach utilizes multi-head graph attention mechanisms [11], [26], which adaptively weight interactions among biomarkers based on both spatial distance and biological significance, thereby enabling a more realistic modeling of the functional interactions among biomarkers. On the other hand, although these models are capable of accurately discovering spatial patterns through machine learning algorithms, they typically require supervised learning, which necessitates large amounts of annotated data. This is costly and would prove infeasible to implement in practice in biomedical imaging in the real world [11], [12], [16]–[18]. Also, these limitations significantly restrict the model’s applicability and scalability across various datasets. In our proposed method, to address this limitation, we utilize self-supervised contrastive learning [7], [13], [27] to directly discover biologically meaningful regions from unlabeled CycIF data, thereby extracting intrinsic spatial and structural information without the need for annotation and using it to identify patterns. Beyond the reliance on labeled data, another fundamental limitation of these approaches lies in their predominantly two-dimensional formulation. Most self-supervision approaches in spatial imaging operate on 2D tissue sections [9], [14], [16], [28], which inherently lose depth information that is critical for understanding the volumetric organization of tissue and tumor-immune interactions, including essential details for biological interpretability and marker correlation. Our model operates on 3D spatial graphs that preserve depth-dependent interactions between biomarkers, enabling the discovery of an ROI that captures the authentic volumetric architecture of the tissue. The lack of effective three-dimensional modeling further impacts how results can be interpreted and validated by domain experts. Existing visual analytics systems, such as histoCAT, ImaCytE, and Visinity, support the interactive exploration of multiplexed images but remain largely 2D-centric and disconnected from learned model outputs [22], [29], [30]. As a result, experts must manually search for dense images, which may lead to the overlooking of salient regions. We fill this gap by integrating

model-detected ROIs into the Vitesse framework [24] and leveraging React components to provide experts with deeper insight into 3D images and attention to biologically important regions, while enabling interactive 3D exploration and validation.

III. METHODOLOGY

Our method defines a 3D computational pipeline for identifying biologically meaningful ROIs in multiplex, high-dimensional CycIF imaging volumes. It uses a 3D graph structure, an attention-based representation network, and self-supervised learning to model biomarkers and identify the most biologically meaningful interactions among them within the volumetric tissue. Each ROI represents a meaningful biological interaction, such as an immune-tumor niche, helping experts make more accurate diagnoses and inform treatment decisions.

A. Graph definition

The objective of this work is to identify biologically meaningful ROIs, such as immune-tumor niches, within high-dimensional CycIF tissue volumes. Each **dataset** is represented as a five-dimensional tensor $V \in \mathbb{R}^{t \times c \times z \times y \times x}$, where $c > 30$ denotes the number of biomarker channels. Each voxel $V(c, z, y, x)$ corresponds to the fluorescence intensity of channel c at spatial location (z, y, x) . The temporal dimension t represents the acquisition time and remains constant in this study.

To capture the functional organization of the tissue microenvironment, the method groups biomarkers into a predefined set of biologically meaningful categories, as shown in Eq. 1. Each category comprises a collection of biologically related biomarkers whose coordinated expression characterizes a specific molecular or cellular microenvironment. In the method, the algorithm processes one or more input categories, which then identifies similar categories within the entire dataset.

$$\mathcal{C} = \{C_1, C_2, C_3, \dots, C_n\} \quad (1)$$

Here, C_k denotes the set of biomarkers associated with category k . For each category, in Eq. 2, intensities are mapped to their corresponding category sets and normalized.

$$I'_c = \text{Norm}(\text{FlatField}(I_c)), \quad c \in C_k, k = 1, \dots, n \quad (2)$$

where $\text{FlatField}(\cdot)$ corrects channel-specific intensity bias and $\text{Norm}(\cdot)$ applies per-channel normalization.

The method partitions the tissue volume into supervoxels, denoted as S_i , where each supervoxel is a spatially contiguous block of $n \times n \times n$ voxels. The parameter n is chosen to match biologically relevant interaction scales within the tissue microenvironment [28], [31]–[33]. This local, signal-centric representation aggregates voxel-level information into semantically coherent units, substantially reducing computational complexity while preserving essential spatial relationships. As a result, it enables scalable construction and efficient analysis of graph-based tissue representations.

$$\mathcal{S} = \{S_1, S_2, \dots, S_n\} \quad (3)$$

Each supervoxel consists of multiple voxels, where each voxel contains zero or more biomarker intensity values. For each supervoxel, biomarker feature descriptors are computed by averaging voxel-level intensities across all voxels contained within the supervoxel. For each node i representing a supervoxel S_i , a set of morphological descriptors $M_i = [V_i, A_i, T_i]$ is extracted, where V_i denotes volume, A_i surface area, and T_i texture features. Each node is additionally associated with a three-dimensional centroid $X_i = (x_i, y_i, z_i)$. Furthermore, per-channel intensity statistics are computed for each biomarker to effectively capture localized signal information.

B. Category-Aware Feature Construction

The approach incorporates functional biomarker groupings by defining biologically meaningful categories, denoted as C_k . Within each category, biomarker characteristics are integrated using a lightweight attention-based multilayer perceptron. Eq. 4 defines the resulting category-specific feature embedding for node i .

$$F_i^{(k)} = \sum_{m \in C_k} \alpha_m^{(k)} \cdot \phi_m(\text{stats}_{i,m}) \quad (4)$$

Here, $\alpha_m^{(k)}$ denotes the learned attention weight for biomarker m within category k , while $\text{stats}_{i,m}$ represents the intensity statistics of biomarker m at node i . The function $\phi_m(\cdot)$ denotes a nonlinear transformation. The resulting node-level feature vector H_i is constructed as defined in Eq. 5.

$$H_i = [F_i^{(1)}, F_i^{(2)}, \dots, F_i^{(K)}, M_i, X_i] \quad (5)$$

Next, the algorithm constructs a three-dimensional spatial graph $G = (V, E)$, where each node $v_i \in V$ represents a supervoxel. An edge is formed between nodes v_i and v_j if the Euclidean distance between their centroids lies within a fixed spatial radius r :

$$\|X_i - X_j\|_2 \leq r \quad (6)$$

Next, in Eq. 7 each edge is assigned a weight W_{ij} that captures both spatial proximity and biological similarity across biomarker categories:

$$W_{ij} = \frac{1}{\|X_i - X_j\|_2 + \varepsilon} + \gamma \cdot \text{sim}\left(\sum_k \beta_k F_i^{(k)}, \sum_k \beta_k F_j^{(k)}\right), \quad (7)$$

Here, β_k controls the relative contribution of each biomarker category, γ balances biological similarity against spatial distance, $\text{sim}(\cdot, \cdot)$ denotes a cosine similarity function, and ε is a small constant introduced for numerical stability. A shared GAT is trained on G to generate node embeddings Z_i . For each category k , a linear classification head then produces a category-specific saliency score using a sigmoid activation function.

Where β_k determines each category's contribution, γ balances biological similarity against spatial distance, $\text{sim}(\cdot, \cdot)$ is a cosine similarity function, and ε equals one for numerical stability. A shared GAT trained on G generates node embeddings Z_i for each category k by sigmoid in Eq. 8; a linear

classification head produces a category-specific saliency score using the sigmoid function.

$$s_i^{(k)} = \sigma(w_k^\top Z_i) \quad (8)$$

where w_k is a vector of learnable weights, w , and $\sigma(\cdot)$ is defined as a sigmoid activation function. High values of $s_i^{(k)}$ indicate that the node i is more locally relevant to category k .

C. Self-Supervised Training Objectives

The GAT can learn and improve its performance through self-supervised techniques, meaning it can optimize its parameters directly without relying on a separate model for training. In our model, GAT exploits the inherent spatial structure and co-expression patterns of biomarkers in CycIF volumes, rather than using manually annotated ROIs. A single GAT encoder is trained simultaneously on all sampled subgraphs, ensuring consistent representation learning across the entire tissue volume. Each training sample is treated as a local subgraph, providing independent information to the network. Additionally, the method continuously updates information about the edges that connect supervoxels, ensuring the learning network method continuously updates the information of the supervoxels and their connections.

a) **Masked-channel recovery:** To improve the learnability of a strong biomarker, each category C_k masks a random subset of biomarker channels at a node. The GAT is trained to recover the masked intensities using information from unmasked channels and neighboring nodes. This approach encourages the creation of redundancy-aware representations, increasing robustness to missing or distorted data.

b) **Spatial distance regression:** involves training the model to explicitly encode geometric relationships by predicting the Euclidean distance between the centroids of nodes based on their learned embeddings. For nodes i and j with centroids X_i and X_j , the target distance is $d_{ij} = \|X_i - X_j\|_2$. This method helps ensure that the embedding space accurately reflects spatial relationships in three dimensions (x, y, z) .

c) **Contrastive consistency:** Contrastive learning uses graph augmentations such as node masking, edge dropping, and subgraph sampling to ensure invariance to local perturbations. This method fosters similar embeddings for augmented versions of the same subgraph while pushing apart embeddings of different subgraphs, thereby stabilizing the learning process.

To capture three-dimensional neighborhood coherence, node embeddings include category-specific saliency scores within a biologically meaningful interaction radius R , which is equal to or greater than the supervoxel size, as shown in Eq. 9 for the spatial neighborhood of a node i .

$$\mathcal{N}_R(i) = \{i \mid \|X_i - X_j\|_2 \leq R\} \quad (9)$$

For each node i and biomarker category k , a radius-based interaction score is computed by RadScore in Eq. 10.

$$\text{RadScore}_R^{(k)}(i) = \frac{\sum_{j \in \mathcal{N}_R(i)} a_{ij} s_j^{(k)}}{\sum_{j \in \mathcal{N}_R(i)} a_{ij} + \varepsilon} \quad (10)$$

where a_{ij} denotes the learned attention coefficient between nodes i and j , $s_j^{(k)}$ is the category-specific saliency score, and ε ensures numerical stability.

Nodes with high $\text{RadScore}_R^{(k)}$ values are grouped into spatially connected components, forming ROIs and for each ROI $R^{(k)}$, a regional biomarker signature, denoted as μ , is computed and defined in Eq. 11.

$$\mu^{(k)} = \frac{1}{|R^{(k)}|} \sum_{i \in R^{(k)}} \mathbf{F}_i^{(k)} \quad (11)$$

Here, $\mathbf{F}_i^{(k)}$ represents the category-specific biomarker embedding for node i . The graph connectivity radius is initialized to r , and the neighborhood aggregation radius is set to R , consistent with prior work on inter- and intra-cellular immune-tumor interactions. These parameters are subsequently refined through validation across the entire dataset.

D. ROI Scoring and Coordinate Extraction

After self-supervised optimization, the trained GAT encoder was used to analyze the tissue volume, score potential spots, and identify seed coordinates for interesting regions. The candidate of each subgraph g_i , located at a specific center with coordinates $\mathbf{X}_i = (x_i, y_i, z_i)$, and the shared GAT encoder will process it entirely through the graphs to create node-level embeddings $\{Z_u\}_{u \in g_i}$. These embeddings capture the local context and form a subgraph-level representation for nodes within each subgraph in each h_i in Eq. 12, and utilize a global pooling operator to enhance the analysis method.

$$\mathbf{h}_i = \text{Pool}(\{\mathbf{Z}_u\}_{u \in g_i}) \quad (12)$$

where $\text{Pool}(\cdot)$ denotes the concatenation of max pooling, capturing complementary aspects of neighborhood structure and feature distribution. A lightweight scoring head $q_\omega(\cdot)$ then maps the pooled representation to a scalar interaction score.

$$\text{Score}(g_i) = q_\omega(\mathbf{h}_i) \quad (13)$$

This assesses the strength and coherence of biologically relevant spatial-molecular interactions within the local three-dimensional space. All candidate centers are ranked in descending order based on their $\text{Score}(g_i)$ in Eq. 14, and the top- K locations are selected as seed points for the ROIs.

$$\mathcal{P}_K = \text{TopK}\left(\{(\mathbf{X}_i, \text{Score}(g_i))\}_{i=1}^M\right) \quad (14)$$

To minimize duplicate detections in densely sampled regions, an optional non-maximum suppression step is used. This step discards candidate centers whose Euclidean distance in (x, y, z) space is less than a specified tolerance radius r_{nms} from a higher-scoring location. The remaining elements of $\mathcal{P}_K$ form the final predicted ROI coordinates, which can be directly overlaid onto the original CycIF volume for expert visualization, validation, and further quantitative analysis.

Algorithm 1

ConGAT: ROI Discovery in Microscopic Imaging

Require: 5D CycIF volume $V \in \mathbb{R}^{T \times C \times Z \times Y \times X}$, graph radius r , interaction radius R , biological weight γ , temperature τ
Ensure: 3D ROI masks $\{\mathcal{R}^{(k)}\}$, biomarker signatures $\{\mu^{(k)}\}$, attention heatmaps
 Map biomarkers to biological categories $\{C_k\}$
 Partition V into 3D supervoxels and compute node features $(\mathbf{X}, \mathbf{M}, \mathbf{F})$
 Construct spatial graph $G = (V, E)$, $d_{ij} = \|\mathbf{X}_i - \mathbf{X}_j\|_2 \leq r$
 Initialize shared GAT backbone and category-specific saliency heads
 Train GAT using self-supervised objectives (masked-channel recovery, spatial regression, contrastive consistency) weighted by γ and temperature τ
 Compute node embeddings $\mathbf{Z}$ and category saliency maps
 $s^{(k)} = \sigma(\mathbf{Z}\mathbf{w}_k)$
 Aggregate saliency within radius R to compute interaction scores $\text{RadScore}_R^{(k)}$
 Threshold top $p\%$ of $\text{RadScore}_R^{(k)}$ and extract connected components as ROIs $\mathcal{R}^{(k)}$
 Summarize each ROI by centroid, volume, mean saliency, and biomarker signature $\mu^{(k)}$
return $\{\mathcal{R}^{(k)}\}$, $\{\mu^{(k)}\}$, attention heatmaps

IV. DATASET

A. Original dataset

The primary dataset of this study is a 3D multiplex microscopy dataset of metastatic melanoma skin tissue generated using CycIF microscopy imaging technology. In this dataset, the spatial and intensity distributions of more than 30 biomarkers are recorded with single-cell resolution in a volumetric sample with dimensions of $194 \times 5508 \times 10908$ at a specific time. The dataset forms a five-dimensional image that encodes spatial, intensity, and temporal information. This dataset displays the pattern of intracellular biomarkers placement next to each other along with their intensity and morphology, and the spatial distance between them, based on which the areas related to immune cells and tumor cells, and their interactions can be identified and studied more precisely, which helps specialists to study the treatment process and the effect of drugs [28], [34], [35].

Specific microenvironments in biology are characterized by a unique set of biological biomarkers, enabling the identification of distinct, meaningful environments based on the presence and proximity of these biomarkers. A great variety of these types of interactions is typically defined in biology, providing researchers with different applications. In this study, we aim to investigate the immune and tumor environments, as well as their interactions, within the dataset. Therefore, we define the following categories, and in our algorithm, we search for environments that include them, highlighting them for experts to pay more attention to and follow up on those in the early stages. In this study, we defined four categories, each representing a compromise of the biomarkers listed in the table I, with one or more potentially included in our dataset. The

classification provides a structured representation of multiple data that enables category-aware modeling and facilitates the interpretation of spatial-marker patterns. Given the massive

TABLE I
REPRESENTATIVE BIOMARKERS FOR TISSUE CHARACTERIZATION.

Category	Representative biomarkers
T-cell Maturation	CD3E, CD4, CD8A, CD45RA, CD45RO, CCR7, CD27, CD28, TCF7
Inflammatory Zone	IFNG, TNF α , IL6, CXCL9, CXCL10, ICAM1, VCAM1, NF- κ B
Oxidative Stress Regulation	COX-IV, Catalase, SOD1, SOD2, NRF2, GPX1, PRDX1
B-cell Infiltration	CD20, CD19, CD79A, CD79B, MS4A1, CD38, MZB1

size of the data, which includes more than eleven billion voxels, processing it is very time-consuming and computationally demanding. Previous studies have demonstrated that coordinated immune activity and short-range signaling occur within localized neighborhoods, approximately 10–30 μ m in diameter. At the same time, larger spatial aggregations risk merging biologically unrelated regions [28], [31]–[33]. We decided to use supervoxels with dimensions of $16 \times 16 \times 16$ voxels, which ultimately result in about three million supervoxels and are very suitable for an algorithm based on the self-supervised GAT.

Supervoxel construction is performed by calculating the average intensity of each biomarker within a supervoxel and placing the supervoxel’s centroid at its spatial location. Supervoxelization helps ensure that the updated dataset will have a biologically significant equilibrium that preserves meaningful spatial interactions while reducing computational requirements.

B. Synthetic Dataset

To further test the strength and generalizability of the model, we considered creating a synthetic dataset that includes all the main characteristics, such as spatial and structural distances of real tissue, and where we injected controlled variations in the size and overlap of biomarkers. Therefore, to create synthetic datasets, we use dimensions equal to those of the real data that have been supervoxelized and attempt to expand them in the X, Y, and Z directions. We also try to simulate different textures for the dataset based on the same real data, where the magnitude of biomarker intensities and spatial distances between them vary, allowing us to also evaluate the ConGAT method in terms of structural variations in the data. This method enables a comprehensive analysis of datasets. Ultimately, it ensures rigorous and biologically relevant validation of the predictive model [36]–[38].

V. EVALUATION

We validated the ConGAT model in two ways to evaluate both the algorithmic performance and the biological relevance considered in its design. In the first method, due to the lack of similar datasets, we created a controlled synthetic dataset

based on information from a known original dataset that has features of different sizes and patterns than the original dataset. We consider these datasets individually each time as input to the ConGAT system. After applying the algorithm to them and comparing the outputs with specific ground truth values, we can evaluate our algorithm. Next, we consulted experts in the fields of biology and visualization to evaluate the model’s outputs. In this evaluation, we asked both groups of experts to identify their areas of interest, which were essential to them in terms of the available information, in the original dataset displayed by the Vitesse framework in an interactive 3D browser, and finally, to record these spatial locations and compare them with the outputs of our ConGAT model.

A. Synthetic Data Evaluation

We provide a controlled quantitative evaluation of the learning dynamics of ConGAT using synthetic five-dimensional data closely resembling the original data, the first. These synthetic data also allow for tight control of imaging properties, such as anisotropic spatial scaling, texture changes, and the enumeration of biomarker complexities. Therefore, multiple synthetic volumes are generated by independently varying scale along the x, y, and z axes, modifying texture patterns to simulate heterogeneous tissue appearance, and altering both the number of biomarkers and the degree of spatial overlap between biomarker channels. The trained ConGAT is trained on synthetic training volumes over 20 epochs and tested on held-out test volumes having explicit ground-truth ROIs. In this arrangement, the objective evaluation of the model’s performance in identifying and localizing interaction-rich areas is feasible [36]. Ground truth is compared with model predictions, and standard classification measures, such as precision, recall, F1 score, and accuracy, are employed to evaluate forecasts against ground truth [39] which are defined in Eq. 15.

$$\begin{aligned} \text{Precision} &= \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \\ \text{F1 Score} &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}, \\ \text{Accuracy} &= \frac{TP + TN}{TP + TN + FP + FN} \end{aligned} \quad (15)$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively [39]. These metrics evaluate the classification performance of the model in identifying interaction-rich regions.

To assess spatial localization accuracy, we further employ the intersection-over-union (IoU) metric, which measures the volumetric overlap between predicted ROIs V_p and ground-truth ROIs V_g .

$$\text{IoU} = \frac{|V_{\text{predicted}} \cap V_{\text{ground truth}}|}{|V_{\text{predicted}} \cup V_{\text{ground truth}}|} \quad (16)$$

The current evaluation systematically varies spatial scale, texture, and biomarker overlap to examine how changes in spatial resolution and visual appearance affect model performance. Together, these metrics provide a detailed assessment

of the robustness of ConGAT and its ability to generalize across diverse data conditions.

B. Expert Evaluation on Real Tissue

Although synthetic benchmarks are necessary to confirm the effectiveness of algorithms, they do not accurately represent the biological complexity of the models that consist of real tissue microenvironments. To overcome this limitation and determine visual and biological fidelity, we conducted an expert-guided test using the original high-dimensional multiplexed CycIF melanoma dataset [40]. Ten domain experts participated in the study, comprising six visualization researchers and four senior cancer biology researchers with extensive experience in studying the tumor microenvironment. Each expert independently examined the volumetric tissue data and identified five key ROIs within each microenvironment. Salient regions were those with high biomarker levels, strong marker co-localization, and biological significance, such as immune niches and immune–tumor interfaces. The experts provided the (x, y, z) coordinates for each ROI, which served as reference points for important biological spatial locations. To provide a biologically based comparison between expert annotations and model predictions, every region of interest is replaced by a three-dimensional volume of a cylinder, with the center being the given coordinate. The cylindrical volume is sliced into 48 cuts along the Z axis, which is equal to three supervoxels, each of voxel height 16, or 24 cuts below and 24 cuts above the central z -coordinate. The axial extent of the data corresponds to a physical extent of 10–30 μm , which is also within the range of immune–tumor interaction ranges and the scale of short-range paracrine signaling, as established in the literature [28], [31]–[33]. This representation describes local volumes of space with three dimensions of neighborhoods, while maintaining biological specificity and eliminating the aggregation of spatially dissimilar tissue volumes.

Model-predicted ROIs generated by ConGAT were evaluated against expert-defined ROIs using two complementary spatial metrics: volumetric overlap and spatial localization accuracy. Volumetric overlap that was quantified using the 3D IoU between the cylindrical volumes V_{model} and V_{expert} calculated by Eq. 16.

This metric assesses the accuracy with which predicted interaction regions align with biologically relevant neighborhoods identified by domain experts, based on their spatial overlap. It quantifies localization precision using the Euclidean distance between the centroids of these regions. For centroid coordinates (x_e, y_e, z_e) and (x_m, y_m, z_m) , the distance is computed by Eq. 17.

$$d = \sqrt{(x_m - x_e)^2 + (y_m - y_e)^2 + (z_m - z_e)^2} \quad (17)$$

Centroid distance measures localization error by considering voxel resolution and expressing deviations in micrometers (μm). Smaller distances indicate better agreement between the model’s predicted regions and those identified by domain experts. When used with IoU, centroid distance provides

a combined assessment of overlap and accuracy, offering a comprehensive view of spatial consistency and biological relevance in ConGAT’s predictions. Together, these metrics highlight the robustness of ConGAT as a self-supervised model for pinpointing biologically relevant interaction regions in complex 3D tumor–immune microenvironments.

VI. RESULT

This section provides a detailed analysis of ConGAT, utilizing a combination of controlled synthetic experiments and expert-based validation on real 3D tissue data. Since there are limitations in the number of reliable voxel-based ground truth annotation datasets, we first turn to carefully designed synthetic datasets that match the dimensional scale and biomarker composition of real volumes. These artificial datasets enable a strict and repeatable evaluation of model accuracy and robustness to systematic changes in spatial scale and textual heterogeneity, as well as the accurate quantification of spatial overlap in volumetric and distance-based measures. We then consider the biological relevance and interpretability of ConGAT on real multiplexed tissue volumes by expert-guided visual analysis by two complementary groups: visualization researchers, who are interested in regions with high marker intensity and spatial concentration, and cancer biologists, who are interested in regions with strong interactions between biologically meaningful biomarkers.

A. Synthetic Data Evaluation

We use synthetic 3D data with known ground truth for controlled assessment. Unlike real biological data, synthetic data provides precise control over spatial and appearance parameters, allowing for a quantitative comparison with reference regions. We create an initial synthetic 3D volume matching the original microscopy data as a ground-truth reference. To test robustness across diverse patterns and distortions, we generate synthetic datasets with geometric and structural variations applied separately along the x , y , and z axes, scaled by a factor of 1 to 3, resulting in ten configurations. We also incorporate biological appearance variability by introducing five distinct synthetic textures that mimic different tissue microenvironments, adjusting biomarkers, intensity distributions, spatial concentrations, and inter-biomarker interactions. These techniques produce multiplexed tissue volumes with exact ROI geometry and similar textures, combining appearance and spatial variability.

For each scale and texture setup, ConGAT detects high signal regions and significant interactions, ranking the top 100 regions based on scores. It evaluates similarity between predicted and actual coordinates using supervoxel classification and the Euclidean distance method. Metrics assess the correspondence between predicted and ground-truth regions:

a) **Volumetric overlap**: measures the proportion of shared volume between two regions, comparing the model-predicted point and the ground-truth point, using a specified radius and height.

b) **Euclidean distance**: measures the three-dimensional space between the predicted center and the actual ground-truth center.

1) **Volumetric Overlap Analysis**: ConGAT performs consistently when volumetric ratios stay within a moderate range, and the overall pattern structure remains intact. As shown in Fig. 1, the model displays significant volumetric overlap across various geometric and textural scaling scenarios, highlighting its ability to adapt to different spatial scales and patterns. However, performance diminishes under more extreme geometric distortions, especially with severe anisotropic scaling. When scaling ratios exceed an order of magnitude along one axis, spatial patterns become increasingly fragmented, which hinders the detection of effective patterns in other dimensions and results in lower accuracy.

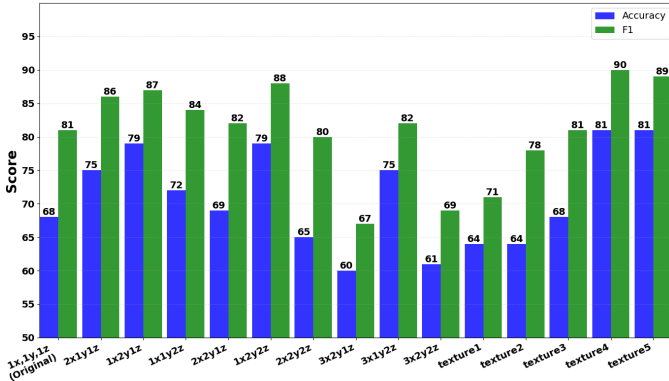


Fig. 1. Evaluation of volumetric overlap: Accuracy and F1 Score across scale and texture configurations.

2) **Euclidean Distance Analysis**: Volumetric overlap measures how much two regions spatially coincide, but it doesn't explicitly evaluate if the predicted areas correspond to biologically or visually significant centers of interest. To assess the accuracy of our localization, we measure the Euclidean distance between the predicted ROI centers from ConGAT and the true ROI centers, mirroring how experts choose regions. This measure assesses the model's accuracy in pinpointing the spatial focus of a region, regardless of the region's shape or volume. Fig. 2 shows centroid-based Euclidean distances measured at the voxel level. In most scale and texture scenarios, ConGAT exhibits low centroid distances, indicating that the predicted ROIs align closely with the ground-truth centers. This suggests accurate center-based detection, which is vital in realistic analysis settings where experts focus on the central location of biologically or visually important regions before further investigation. In this case, we also observe that extreme anisotropic scaling and specific patterns have a negative impact on model accuracy in determining the accurate centroid. During this assessment, we find that ConGAT consistently produces reliable results across different synthetic textures and scales, showing minimal sensitivity to variations in texture and size. The high volumetric overlap, along with a low Euclidean distance, indicates accurate regional localization and a solid understanding of spatial relationships. These metrics

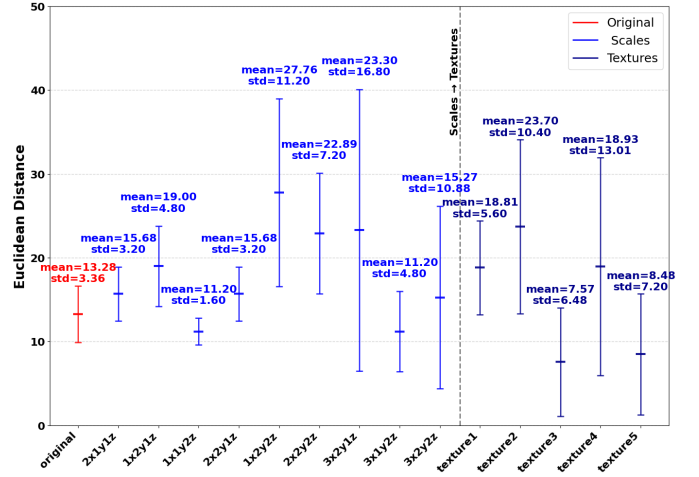


Fig. 2. Voxel-level centroid Euclidean distances showing close ROI-ground-truth alignment.

together give a thorough evaluation of spatial accuracy: overlap reflects regional agreement, while Euclidean distance assesses localization precision. In general, the evidence suggests that ConGAT is effective outside of synthetic data and can be trusted to apply to real 3D microscopic data, indicating its ability to accurately capture intricate spatial relationships even when multiple textures and distortions are present.

B. Experts' Validation

We evaluated the biological relevance and interpretability of regions identified by ConGAT through an expert-guided validation study using the original multi-dimensional CycIF melanoma tissue volume [41]. The objective of this study was to determine whether model-detected regions correspond to biologically meaningful tissue microenvironments as recognized by human experts. We engaged ten domain experts, including six computer science researchers from the University of Illinois Chicago with varying levels of experience in biomedical data visualization and four cancer biology researchers from the Department of Cell Biology at Rush Medical University. This interdisciplinary composition allowed us to integrate both perceptual and biological expertise when assessing the validity of the model's spatial predictions.

Before the evaluation, we provided visualization experts with explicit guidance on how to interpret the data and define an interesting region. In particular, we emphasized regions characterized by high biomarker intensity and intense spatial concentration of signal within localized areas. We then introduced Vitessce [24], an interactive and colorful three-dimensional visualization framework that supports simultaneous exploration of multiple biomarker channels. Using this framework, visualization experts examined the entire tissue image and visually integrated patterns across channels. They independently selected their five most interesting regions based on intensity distributions, spatial organization, and visual salience. We followed a complementary protocol for cancer biology experts. After explaining the visualization framework,

we demonstrated how to interact with the 3D tissue volume, including rotating, filtering intensity, and dynamically adding or removing biomarker channels. Biology experts, focusing on biologically relevant microenvironments informed by their domain knowledge, identified five regions they considered most biologically meaningful for validation.

To quantify agreement between expert judgment and model predictions, we compared expert-selected regions with ConGAT-detected regions using complementary spatial metrics. For each microenvironment category, we reported the top five ConGAT-detected regions with the highest confidence, defined by the model’s learned interaction strength. Fig. 3 shows a representative sample from one selected microenvironment, labeled the Inflammatory Zone. In this figure, colored markers indicate regions selected by experts as interesting, while filled black circles denote regions identified by ConGAT. The close spatial proximity between expert selections and model detections indicates strong agreement in spatial distance, providing qualitative evidence that ConGAT captures biologically meaningful spatial patterns consistent with prior observations of cellular interactions in multiplexed tissue imaging [4], [29]. We

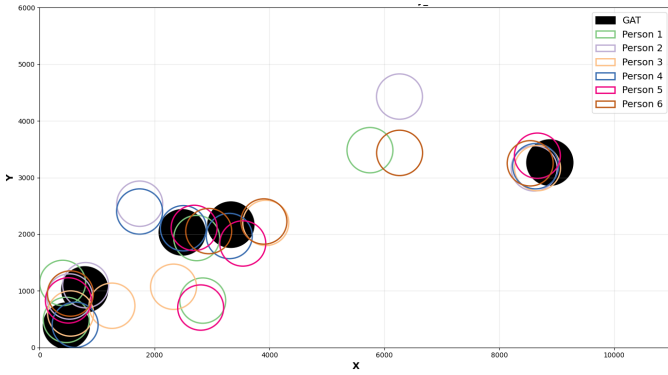


Fig. 3. Representative 2D illustrating expert-selected regions and ConGAT detections in the Inflammatory Zone.

quantified the correspondence between expert-selected regions and ConGAT predictions using volumetric overlap analysis. Fig. 4 shows inter-rater plots of intersection-over-union (IoU) scores across microenvironment categories and experts, where columns represent individual experts, rows denote microenvironments, and the final column aggregates results across all experts. The aggregated IoU scores range from 40% to 80%, indicating consistent volumetric overlap and strong regional-level agreement between expert annotations and the highest-ranked regions detected by ConGAT. In addition to volumetric overlap, we evaluated spatial localization accuracy by computing the Euclidean distance between expert-selected coordinates and the centroids of regions predicted by ConGAT. Fig. 5 presents inter-rater error bar plots of these distances across experts and microenvironment categories, highlighting the distribution and variability of spatial deviations in physical units. Lower mean distances indicate stronger spatial correspondence between expert annotations and model predictions. Complementing this analysis, Fig. 6 presents a heatmap of Euclidean

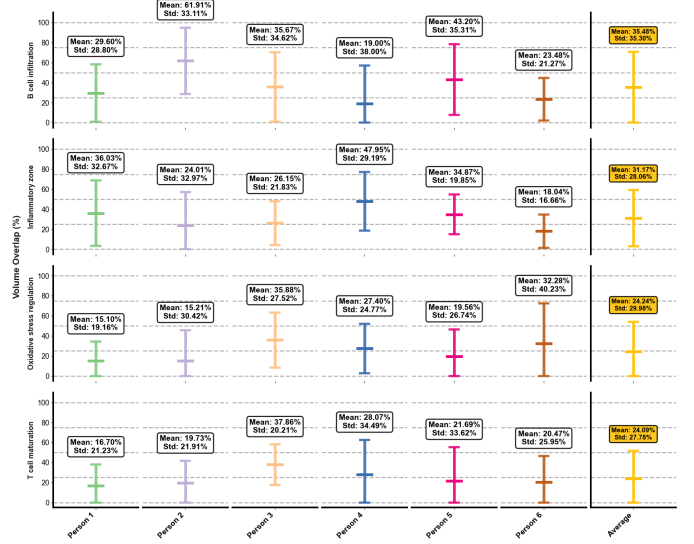


Fig. 4. Inter-rater of 3D overlap between ConGAT ROIs and expert-selected ROIs across participants and microenvironments.

distances across experts and microenvironments, facilitating the identification of systematic trends and microenvironment-dependent differences in localization accuracy.

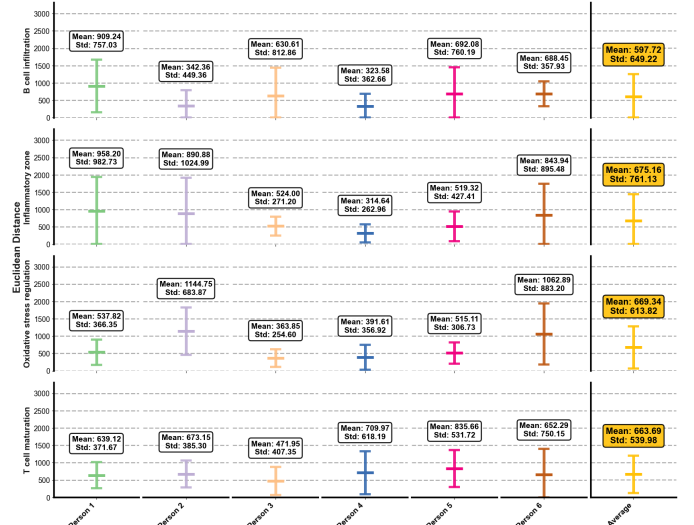


Fig. 5. Inter-rater Euclidean distance error bars between expert-selected regions and ConGAT predictions.

Taken together, the expert-guided validation demonstrates that ConGAT robustly identifies biologically meaningful ROIs in real three-dimensional multiplexed tissue data. The model exhibits strong agreement with expert-selected regions in both volumetric extent and spatial localization, despite operating without manual annotations. When considered alongside controlled synthetic experiments, which show stable performance across variations in spatial scale and texture, these results confirm that ConGAT effectively captures salient spatial interaction patterns and produces interpretable, expert-aligned

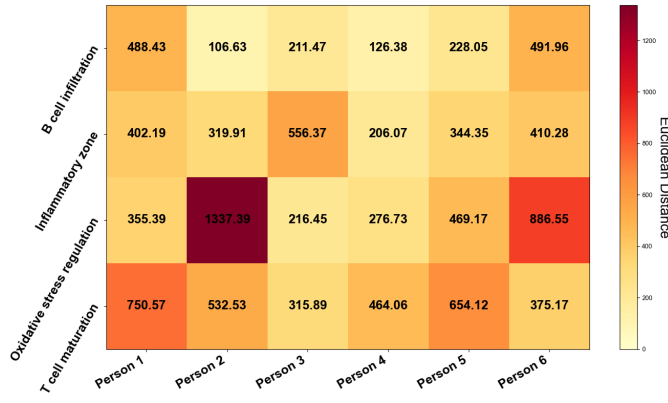


Fig. 6. Median Euclidean distance heatmap between expert annotations and ConGAT predictions.

region detections in complex 3D biological datasets.

VII. DISCUSSION AND CONCLUSION

Multiplexed 3D microscopy imaging provides comprehensive insights into tissue microenvironments, enabling experts to diagnose and treat diseases more effectively. However, these images present analytical challenges for experts to understand all the changes in the entire image due to their high dimensionality, noise distortion, spatial and sample variability, and the scarcity of dependable 3D annotations [9], [31]–[33]. The ConGAT goal addresses these challenges by representing tissues as a 3D graph, where entities are modeled as nodes and links in multidimensional space. Additionally, we leverage category-aware biomarker embeddings and learn meaningful interactions from images using self-supervised methods. We also utilize graph attention mechanisms that emphasize biologically relevant interactions and reduce noise. Controlled synthetic experiments show that ConGAT remains effective across different spatial scales and under texture variations, achieving high volumetric overlap and low centroid localization errors. Its accuracy is not perfect under severe anisotropic distortions, where the local neighborhood geometry becomes less reliable. However, we propose solutions to these issues in future research directions. Expert validation using real melanoma CycIF data confirms that ConGAT accurately identifies regions that closely match those recognized by visualization scientists and cancer biologists, as measured by Euclidean distance and overlap metrics, even though it operates without ROI annotations. Integrating ConGAT results into an interactive 3D visualization platform enhances interpretability and supports expert-led exploration and hypothesis generation.

The model’s performance is influenced by the chosen supervoxel resolution and neighborhood radius, assumes well-calibrated spatial geometry, and depends on domain-specific biomarker categories. It has some limitations, which can be mitigated by acquiring more data to develop a more precise machine learning model. Future efforts will include weak supervision, extra self-supervised objectives, and wider validation across various tissues and imaging techniques to enhance

robustness and generalizability.

In conclusion, ConGAT provides a self-supervised graph attention model for identifying biologically relevant regions in high-dimensional multiplexed microscopy imaging. ConGAT addresses the limitations of current methodologies by effectively integrating volumetric structure, biomarker co-expression, and morphological context within a unified graph representation. This innovative approach overcomes the annotation dependencies and two-dimensional constraints of existing techniques. Comprehensive synthetic evaluations illustrate its reliable performance across diverse scales, textures, and complexities of biomarkers. Furthermore, expert studies validate its robust alignment with established tumor–immune microenvironments. Enhanced by interactive 3D visualization, ConGAT establishes a scalable, interpretable, and extensible foundation for spatial tissue analysis and hypothesis-driven exploration in computational pathology.

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